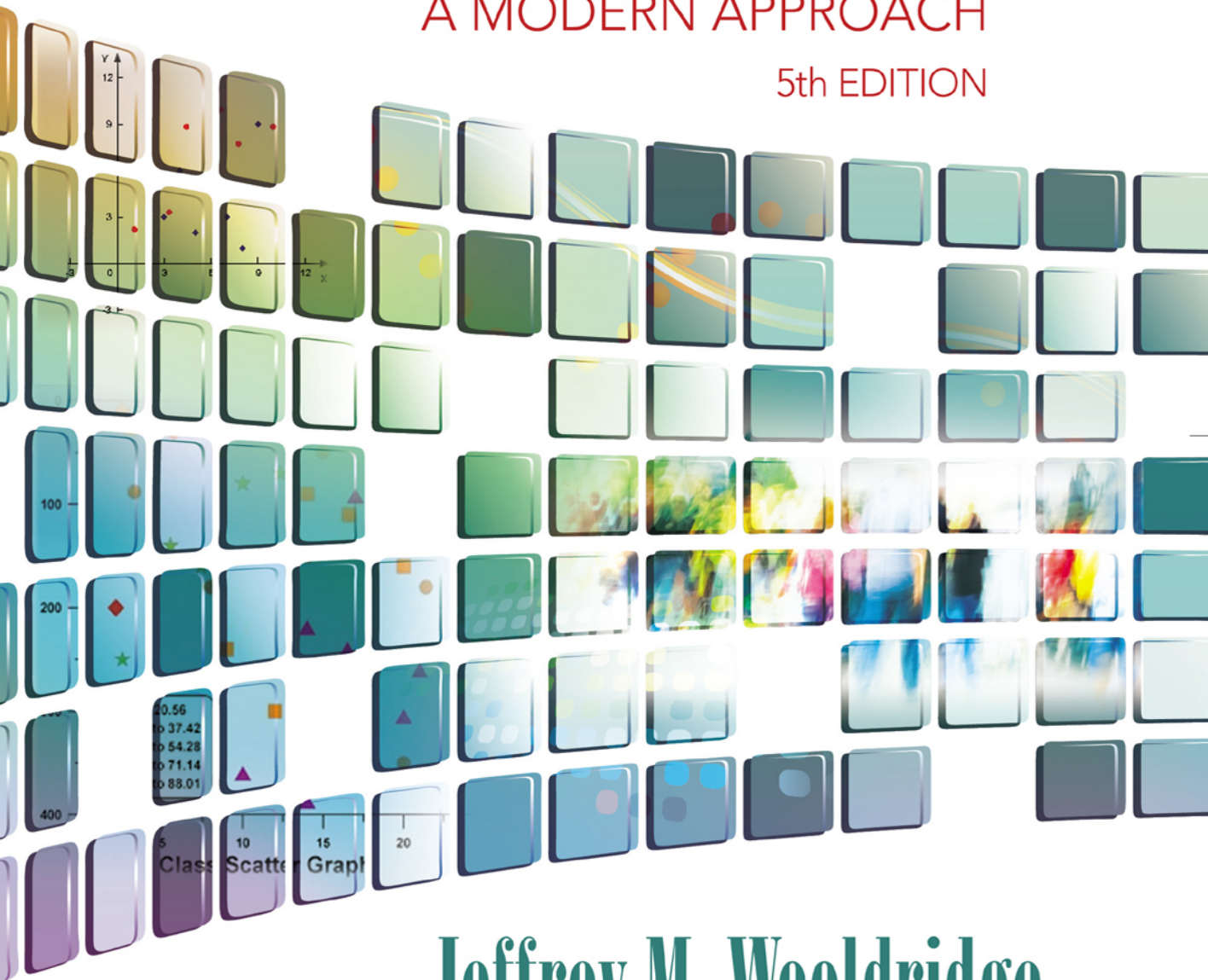


INTRODUCTORY Econometrics

A MODERN APPROACH

5th EDITION



Jeffrey M. Wooldridge



Introductory Econometrics



Introductory Econometrics

A MODERN APPROACH

FIFTH EDITION

Jeffrey M. Wooldridge
Michigan State University



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PREFACE

My motivation for writing the first edition of *Introductory Econometrics: A Modern Approach* was that I saw a fairly wide gap between how econometrics is taught to undergraduates and how empirical researchers think about and apply econometric methods. I became convinced that teaching introductory econometrics from the perspective of professional users of econometrics would actually simplify the presentation, in addition to making the subject much more interesting.

Based on the positive reactions to earlier editions, it appears that my hunch was correct. Many instructors, having a variety of backgrounds and interests and teaching students with different levels of preparation, have embraced the modern approach to econometrics espoused in this text. The emphasis in this edition is still on applying econometrics to real-world problems. Each econometric method is motivated by a particular issue facing researchers analyzing nonexperimental data. The focus in the main text is on understanding and interpreting the assumptions in light of actual empirical applications: the mathematics required is no more than college algebra and basic probability and statistics.

Organized for Today's Econometrics Instructor

The fifth edition preserves the overall organization of the fourth. The most noticeable feature that distinguishes this text from most others is the separation of topics by the kind of data being analyzed. This is a clear departure from the traditional approach, which presents a linear model, lists all assumptions that may be needed at some future point in the analysis, and then proves or asserts results without clearly connecting them to the assumptions. My approach is first to treat, in Part 1, multiple regression analysis with cross-sectional data, under the assumption of random sampling. This setting is natural to students because they are familiar with random sampling from a population in their introductory statistics courses. Importantly, it allows us to distinguish assumptions made about the underlying population regression model—assumptions that can be given economic or behavioral content—from assumptions about how the data were sampled. Discussions about the consequences of nonrandom sampling can be treated in an intuitive fashion after the students have a good grasp of the multiple regression model estimated using random samples.

An important feature of a modern approach is that the explanatory variables—along with the dependent variable—are treated as outcomes of random variables. For the social sciences, allowing random explanatory variables is much more realistic than the traditional assumption of nonrandom explanatory variables. As a nontrivial benefit, the population model/random sampling approach reduces the number of assumptions that students must

absorb and understand. Ironically, the classical approach to regression analysis, which treats the explanatory variables as fixed in repeated samples and is still pervasive in introductory texts, literally applies to data collected in an experimental setting. In addition, the contortions required to state and explain assumptions can be confusing to students.

My focus on the population model emphasizes that the fundamental assumptions underlying regression analysis, such as the zero mean assumption on the unobservable error term, are properly stated conditional on the explanatory variables. This leads to a clear understanding of the kinds of problems, such as heteroskedasticity (nonconstant variance), that can invalidate standard inference procedures. By focusing on the population I am also able to dispel several misconceptions that arise in econometrics texts at all levels. For example, I explain why the usual R -squared is still valid as a goodness-of-fit measure in the presence of heteroskedasticity (Chapter 8) or serially correlated errors (Chapter 12); I provide a simple demonstration that tests for functional form should not be viewed as general tests of omitted variables (Chapter 9); and I explain why one should always include in a regression model extra control variables that are uncorrelated with the explanatory variable of interest, which is often a key policy variable (Chapter 6).

Because the assumptions for cross-sectional analysis are relatively straightforward yet realistic, students can get involved early with serious cross-sectional applications without having to worry about the thorny issues of trends, seasonality, serial correlation, high persistence, and spurious regression that are ubiquitous in time series regression models. Initially, I figured that my treatment of regression with cross-sectional data followed by regression with time series data would find favor with instructors whose own research interests are in applied microeconomics, and that appears to be the case. It has been gratifying that adopters of the text with an applied time series bent have been equally enthusiastic about the structure of the text. By postponing the econometric analysis of time series data, I am able to put proper focus on the potential pitfalls in analyzing time series data that do not arise with cross-sectional data. In effect, time series econometrics finally gets the serious treatment it deserves in an introductory text.

As in the earlier editions, I have consciously chosen topics that are important for reading journal articles and for conducting basic empirical research. Within each topic, I have deliberately omitted many tests and estimation procedures that, while traditionally included in textbooks, have not withstood the empirical test of time. Likewise, I have emphasized more recent topics that have clearly demonstrated their usefulness, such as obtaining test statistics that are robust to heteroskedasticity (or serial correlation) of unknown form, using multiple years of data for policy analysis, or solving the omitted variable problem by instrumental variables methods. I appear to have made fairly good choices, as I have received only a handful of suggestions for adding or deleting material.

I take a systematic approach throughout the text, by which I mean that each topic is presented by building on the previous material in a logical fashion, and assumptions are introduced only as they are needed to obtain a conclusion. For example, empirical researchers who use econometrics in their research understand that not all of the Gauss-Markov assumptions are needed to show that the ordinary least squares (OLS) estimators are unbiased. Yet the vast majority of econometrics texts introduce a complete set of assumptions (many of which are redundant or in some cases even logically conflicting) before proving the unbiasedness of OLS. Similarly, the normality assumption is often included among the assumptions that are needed for the Gauss-Markov Theorem, even though it is fairly well known that normality plays no role in showing that the OLS estimators are the best linear unbiased estimators.

My systematic approach is illustrated by the order of assumptions that I use for multiple regression in Part 1. This structure results in a natural progression for briefly summarizing the role of each assumption:

MLR.1: Introduce the population model and interpret the population parameters (which we hope to estimate).

MLR.2: Introduce random sampling from the population and describe the data that we use to estimate the population parameters.

MLR.3: Add the assumption on the explanatory variables that allows us to compute the estimates from our sample; this is the so-called no perfect collinearity assumption.

MLR.4: Assume that, in the population, the mean of the unobservable error does not depend on the values of the explanatory variables; this is the “mean independence” assumption combined with a zero population mean for the error, and it is the key assumption that delivers unbiasedness of OLS.

After introducing Assumptions MLR.1 to MLR.3, one can discuss the algebraic properties of ordinary least squares—that is, the properties of OLS for a particular set of data. By adding Assumption MLR.4, we can show that OLS is unbiased (and consistent). Assumption MLR.5 (homoskedasticity) is added for the Gauss-Markov Theorem and for the usual OLS variance formulas to be valid. Assumption MLR.6 (normality), which is not introduced until Chapter 4, is added to round out the classical linear model assumptions. The six assumptions are used to obtain exact statistical inference and to conclude that the OLS estimators have the smallest variances among all unbiased estimators.

I use parallel approaches when I turn to the study of large-sample properties and when I treat regression for time series data in Part 2. The careful presentation and discussion of assumptions makes it relatively easy to transition to Part 3, which covers advanced topics that include using pooled cross-sectional data, exploiting panel data structures, and applying instrumental variables methods. Generally, I have strived to provide a unified view of econometrics, where all estimators and test statistics are obtained using just a few intuitively reasonable principles of estimation and testing (which, of course, also have rigorous justification). For example, regression-based tests for heteroskedasticity and serial correlation are easy for students to grasp because they already have a solid understanding of regression. This is in contrast to treatments that give a set of disjointed recipes for out-dated econometric testing procedures.

Throughout the text, I emphasize *ceteris paribus* relationships, which is why, after one chapter on the simple regression model, I move to multiple regression analysis. The multiple regression setting motivates students to think about serious applications early. I also give prominence to policy analysis with all kinds of data structures. Practical topics, such as using proxy variables to obtain *ceteris paribus* effects and interpreting partial effects in models with interaction terms, are covered in a simple fashion.

New to This Edition

I have added new exercises to nearly every chapter. Some are computer exercises using existing data sets, some use new data sets, and others involve using computer simulations to study the properties of the OLS estimator. I have also added more challenging problems that require derivations.

Some of the changes to the text are worth highlighting. In Chapter 3 I have further expanded the discussion of multicollinearity and variance inflation factors, which I first introduced in the fourth edition. Also in Chapter 3 is a new section on the language that researchers should use when discussing equations estimated by ordinary least squares. It is important for beginners to understand the difference between a model and an estimation method and to remember this distinction as they learn about more sophisticated procedures and mature into empirical researchers.

Chapter 5 now includes a more intuitive discussion about how one should think about large-sample analysis, and emphasizes that it is the distribution of sample averages that changes with the sample size; population distributions, by definition, are unchanging. Chapter 6, in addition to providing more discussion of the logarithmic transformation as applied to proportions, now includes a comprehensive list of considerations when using the most common functional forms: logarithms, quadratics, and interaction terms.

Two important additions occur in Chapter 7. First, I clarify how one uses the sum of squared residual F test to obtain the Chow test when the null hypothesis allows an intercept difference across the groups. Second, I have added Section 7.7, which provides a simple yet general discussion of how to interpret linear models when the dependent variable is a discrete response.

Chapter 9 includes more discussion of using proxy variables to account for omitted, confounding factors in multiple regression analysis. My hope is that it dispels some misunderstandings about the purpose of adding proxy variables and the nature of the resulting multicollinearity. In this chapter I have also expanded the discussion of least absolute deviations estimation (LAD). New problems—one about detecting omitted variables bias and one about heteroskedasticity and LAD estimation—have been added to Chapter 9; these should be a good challenge for well-prepared students.

The appendix to Chapter 13 now includes a discussion of standard errors that are robust to both serial correlation and heteroskedasticity in the context of first-differencing estimation with panel data. Such standard errors are computed routinely now in applied microeconomic studies employing panel data methods. A discussion of the theory is beyond the scope of this text but the basic idea is easy to describe. The appendix in Chapter 14 contains a similar discussion for random effects and fixed effects estimation. Chapter 14 also contains a new Section 14.3, which introduces the reader to the “correlated random effects” approach to panel data models with unobserved heterogeneity. While this topic is more advanced, it provides a synthesis of random and fixed effects methods, and leads to important specification tests that are often reported in empirical research.

Chapter 15, on instrumental variables estimation, has been expanded in several ways. The new material includes a warning about checking the signs of coefficients on instrumental variables in reduced form equations, a discussion of how to interpret the reduced form for the dependent variable, and—as with the case of OLS in Chapter 3—emphasizes that instrumental variables is an estimation method, not a “model.”

Targeted at Undergraduates, Adaptable for Master's Students

The text is designed for undergraduate economics majors who have taken college algebra and one semester of introductory probability and statistics. (Appendices A, B, and C contain the requisite background material.) A one-semester or one-quarter econometrics course would not be expected to cover all, or even any, of the more advanced material in Part 3. A typical introductory

course includes Chapters 1 through 8, which cover the basics of simple and multiple regression for cross-sectional data. Provided the emphasis is on intuition and interpreting the empirical examples, the material from the first eight chapters should be accessible to undergraduates in most economics departments. Most instructors will also want to cover at least parts of the chapters on regression analysis with time series data, Chapters 10, 11, and 12, in varying degrees of depth. In the one-semester course that I teach at Michigan State, I cover Chapter 10 fairly carefully, give an overview of the material in Chapter 11, and cover the material on serial correlation in Chapter 12. I find that this basic one-semester course puts students on a solid footing to write empirical papers, such as a term paper, a senior seminar paper, or a senior thesis. Chapter 9 contains more specialized topics that arise in analyzing cross-sectional data, including data problems such as outliers and nonrandom sampling; for a one-semester course, it can be skipped without loss of continuity.

The structure of the text makes it ideal for a course with a cross-sectional or policy analysis focus: the time series chapters can be skipped in lieu of topics from Chapters 9, 13, 14, or 15. Chapter 13 is advanced only in the sense that it treats two new data structures: independently pooled cross sections and two-period panel data analysis. Such data structures are especially useful for policy analysis, and the chapter provides several examples. Students with a good grasp of Chapters 1 through 8 will have little difficulty with Chapter 13. Chapter 14 covers more advanced panel data methods and would probably be covered only in a second course. A good way to end a course on cross-sectional methods is to cover the rudiments of instrumental variables estimation in Chapter 15.

I have used selected material in Part 3, including Chapters 13, 14, 15, and 17, in a senior seminar geared to producing a serious research paper. Along with the basic one-semester course, students who have been exposed to basic panel data analysis, instrumental variables estimation, and limited dependent variable models are in a position to read large segments of the applied social sciences literature. Chapter 17 provides an introduction to the most common limited dependent variable models.

The text is also well suited for an introductory master's level course, where the emphasis is on applications rather than on derivations using matrix algebra. Several instructors have used the text to teach policy analysis at the master's level. For instructors wanting to present the material in matrix form, Appendices D and E are self-contained treatments of the matrix algebra and the multiple regression model in matrix form.

At Michigan State, PhD students in many fields that require data analysis—including accounting, agricultural economics, development economics, economics of education, finance, international economics, labor economics, macroeconomics, political science, and public finance—have found the text to be a useful bridge between the empirical work that they read and the more theoretical econometrics they learn at the PhD level.

Design Features

Numerous in-text questions are scattered throughout, with answers supplied in Appendix F. These questions are intended to provide students with immediate feedback. Each chapter contains many numbered examples. Several of these are case studies drawn from recently published papers, but where I have used my judgment to simplify the analysis, hopefully without sacrificing the main point.

The end-of-chapter problems and computer exercises are heavily oriented toward empirical work, rather than complicated derivations. The students are asked to reason

carefully based on what they have learned. The computer exercises often expand on the in-text examples. Several exercises use data sets from published works or similar data sets that are motivated by published research in economics and other fields.

A pioneering feature of this introductory econometrics text is the extensive glossary. The short definitions and descriptions are a helpful refresher for students studying for exams or reading empirical research that uses econometric methods. I have added and updated several entries for the fifth edition.

Data Sets—Available in Six Formats

This edition adds R data set as an additional format for viewing and analyzing data. In response to popular demand, this edition also provides the Minitab[®] format. With more than 100 data sets in six different formats, including Stata[®], EViews[®], Minitab[®], Microsoft[®] Excel, R, and TeX, the instructor has many options for problem sets, examples, and term projects. Because most of the data sets come from actual research, some are very large. Except for partial lists of data sets to illustrate the various data structures, the data sets are not reported in the text. This book is geared to a course where computer work plays an integral role.

Updated Data Sets Handbook

An extensive data description manual is also available online. This manual contains a list of data sources along with suggestions for ways to use the data sets that are not described in the text. This unique handbook, created by author Jeffrey M. Wooldridge, lists the source of all data sets for quick reference and how each might be used. Because the data book contains page numbers, it is easy to see how the author used the data in the text. Students may want to view the descriptions of each data set and it can help guide instructors in generating new homework exercises, exam problems or term projects. The author also provides suggestions on improving the data sets in this detailed resource that is available on the book's companion website at <http://login.cengage.com> and students can access it free at www.cengagebrain.com.

Instructor Supplements

Instructor's Manual with Solutions

The *Instructor's Manual with Solutions* (978-1-111-57757-5) contains answers to all problems and exercises, as well as teaching tips on how to present the material in each chapter. The instructor's manual also contains sources for each of the data files, with many suggestions for how to use them on problem sets, exams, and term papers. This supplement is available online only to instructors at <http://login.cengage.com>.

PowerPoint slides

Exceptional new PowerPoint[®] presentation slides, created specifically for this edition, help you create engaging, memorable lectures. You'll find teaching slides for each chapter in this edition, including the advanced chapters in Part 3. You can modify or customize the slides for your specific course. PowerPoint[®] slides are available for convenient download on the instructor-only, password-protected portion of the book's companion website at <http://login.cengage.com>.

Scientific Word Slides

Developed by the author, new Scientific Word® slides offer an alternative format for instructors who prefer the Scientific Word® platform, the word processor created by MacKichan Software, Inc. for composing mathematical and technical documents using LaTeX typesetting. These slides are based on the author's actual lectures and are available in PDF and TeX formats for convenient download on the instructor-only, password-protected section of the book's companion website at <http://login.cengage.com>.

Test Bank

In response to user requests, this edition offers a brand new Test Bank written by the author to ensure the highest quality and correspondence with the text. The author has created Test Bank questions from actual tests developed for his own courses. You will find a wealth and variety of problems, ranging from multiple-choice, to questions that require simple statistical derivations to questions that require interpreting computer output. The Test Bank is available for convenient download on the instructor-only, password-protected portion of the companion website at <http://login.cengage.com>.

Student Supplements

The *Student Solutions Manual* contains suggestions on how to read each chapter as well as answers to selected problems and computer exercises. The *Student Solutions Manual* can be purchased as a Printed Access Code (978-1-111-57694-3) or as an Instant Access Code (978-1-111-57693-6) and accessed online at www.cengagebrain.com.

Suggestions for Designing Your Course

I have already commented on the contents of most of the chapters as well as possible outlines for courses. Here I provide more specific comments about material in chapters that might be covered or skipped:

Chapter 9 has some interesting examples (such as a wage regression that includes IQ score as an explanatory variable). The rubric of proxy variables does not have to be formally introduced to present these kinds of examples, and I typically do so when finishing up cross-sectional analysis. In Chapter 12, for a one-semester course, I skip the material on serial correlation robust inference for ordinary least squares as well as dynamic models of heteroskedasticity.

Even in a second course I tend to spend only a little time on Chapter 16, which covers simultaneous equations analysis. I have found that instructors differ widely in their opinions on the importance of teaching simultaneous equations models to undergraduates. Some think this material is fundamental; others think it is rarely applicable. My own view is that simultaneous equations models are overused (see Chapter 16 for a discussion). If one reads applications carefully, omitted variables and measurement error are much more likely to be the reason one adopts instrumental variables estimation, and this is why I use omitted variables to motivate instrumental variables estimation in Chapter 15.

Still, simultaneous equations models are indispensable for estimating demand and supply functions, and they apply in some other important cases as well.

Chapter 17 is the only chapter that considers models inherently nonlinear in their parameters, and this puts an extra burden on the student. The first material one should cover in this chapter is on probit and logit models for binary response. My presentation of Tobit models and censored regression still appears to be novel in introductory texts. I explicitly recognize that the Tobit model is applied to corner solution outcomes on random samples, while censored regression is applied when the data collection process censors the dependent variable at essentially arbitrary thresholds.

Chapter 18 covers some recent important topics from time series econometrics, including testing for unit roots and cointegration. I cover this material only in a second-semester course at either the undergraduate or master's level. A fairly detailed introduction to forecasting is also included in Chapter 18.

Chapter 19, which would be added to the syllabus for a course that requires a term paper, is much more extensive than similar chapters in other texts. It summarizes some of the methods appropriate for various kinds of problems and data structures, points out potential pitfalls, explains in some detail how to write a term paper in empirical economics, and includes suggestions for possible projects.

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Some of the changes I discussed earlier were driven by comments I received from people on this list, and I continue to mull over other specific suggestions made by one or more reviewers.

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Jeffrey M. Wooldridge



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